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Software Architecture Document

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1. Introduction and goals

NOVA converts a natural-language software requirement into a structured test plan, and retains the user's corrections as durable memory so that later work does not repeat the same errors.

Generation is not the architecturally interesting part. The design problem is the loop around it:

1. Converting a freeform correction into a durable, retrievable rule without creating a persistent prompt-injection vector.
2. Retrieving the right rules without an ever-growing prompt that degrades silently.
3. Establishing whether any of it works.

1.1 Requirements overview

The authoritative requirement set is NOVA-SRS-001. The architecturally significant subset:

ID	Requirement	Architectural consequence
FR-2	Schema-conformant generation via constrained decoding	Schema is a first-class shared artifact, not a post-hoc validator

ID	Requirement	Architectural consequence
FR-4	Scored retrieval with scores exposed	Retrieval must be explainable, not merely correct
FR-6.3	No memory stored without explicit confirmation	A human gate sits on the highest-risk write path
FR-8	Complete execution trace, including on failure	Tracing is a structural concern, written incrementally
FR-10.1	Configuration activation requires a passing evaluation	Enforced as a data-layer constraint
NFR-16	Tenant scoping applied at the data access layer	Isolation is structural, not a per-caller convention
NFR-20	The model has no file, command, or network capability	The absence of a tool surface is a designed property
NFR-2	Task duration measured in tens of seconds	Execution is asynchronous and durable

1.2 Quality goals

Ordered. Where two conflict, the higher-ranked prevails.

Rank	Quality goal	Motivation	Scenario reference
1	**Security of the memory path**	A confirmed memory is replayed into every future prompt that retrieves it. Compromise is persistent, not transient	QS-1, QS-2
2	**Analysability**	The value of the system depends on being able to show why an output was produced, and on separating retrieval failure from generation failure	QS-3, QS-4
3	**Testability**	Improvement claims are only meaningful if they are measurable and gated	QS-5, QS-6
4	**Confidentiality**	Requirements and test data must not leave the host in the default configuration	QS-7
5	**Reliability**	A task must not vanish, hang, or lose its trace on failure	QS-8, QS-9

Performance is deliberately absent from this list. Measurement established substantial headroom against the stated target, so it does not currently constrain design.

1.3 Stakeholders

Stakeholder	Concern	Consumed views
Implementing engineer	Buildability, decision rationale	All
Technical reviewer	Engineering judgment, measurement rigour	4, 8, 9, 10, 11

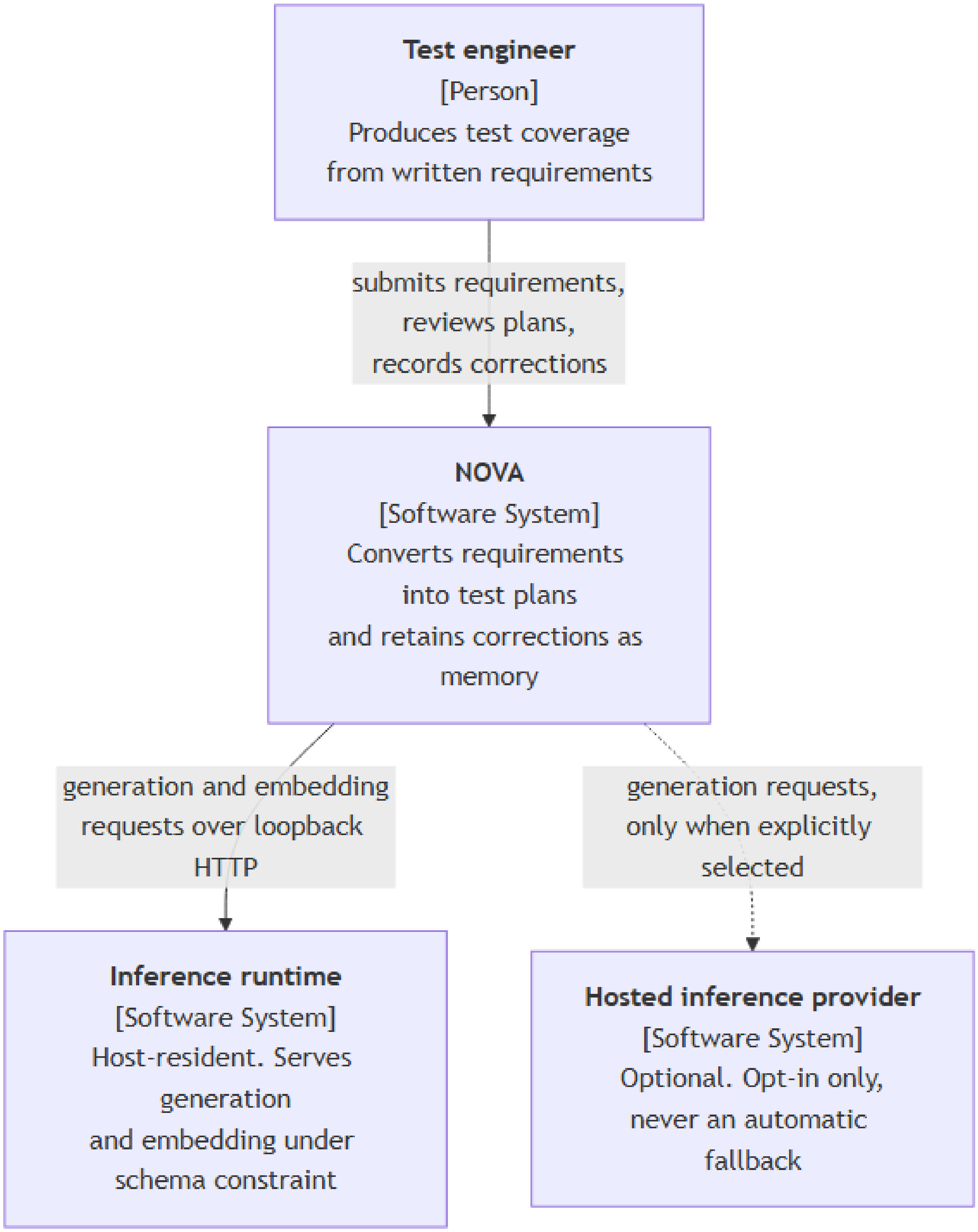
Stakeholder	Concern	Consumed views
Operator	Deployment, failure diagnosis	7, 8
Primary user	Explainability of output	6, 8

2. Architecture constraints

ID	Constraint	Consequence
CON-1	Approximately 130 to 180 engineering hours	One datastore, no framework, no infrastructure that is not load-bearing
CON-2	Single engineer, no external review	Automated gates substitute for peer review where they can
CON-3	Local inference within a 6 GB VRAM budget	Structured-output reliability becomes a primary engineering problem. Model class is 4B parameters, not 8 to 9B
CON-4	Local deployment only	Container composition is the deployment artifact. Inference runs on the host, not in a container
CON-5	Synthetic and open-source data only	Evaluation data is committable. Privacy controls are designed and tested rather than compliance-bound
CON-6	No model training	All adaptation occurs through retrieval and configuration, not weights
CON-7	Single user, multi-tenant-capable schema	Tenant identifier present from the first migration; authentication deferred

3. Context and scope

3.1 Business context (C4 level 1)



External entity	Interface	Notes
Test engineer	Browser, HTTP	The only human actor. No administrative role exists
Inference runtime	HTTP on the host loopback interface	Supplies a JSON Schema per request; returns a conformant completion plus token and timing telemetry
Hosted inference provider	HTTPS	Optional. Selecting it is the only path by which data leaves the host, and it requires an explicit user action

3.2 Technical context

Channel	Protocol	Payload	Trust
Browser to API	HTTP/1.1, JSON	Requirements, feedback, memory operations	Untrusted input
API to datastore	PostgreSQL wire protocol	All persistent state	Internal
Worker to inference runtime	HTTP/1.1, JSON with JSON Schema	Prompts and completions	Output untrusted
API to inference runtime	HTTP/1.1, JSON	Extraction and embedding requests	Output untrusted

Services bind to the loopback interface. The system has no inbound network surface beyond the host.

4. Solution strategy

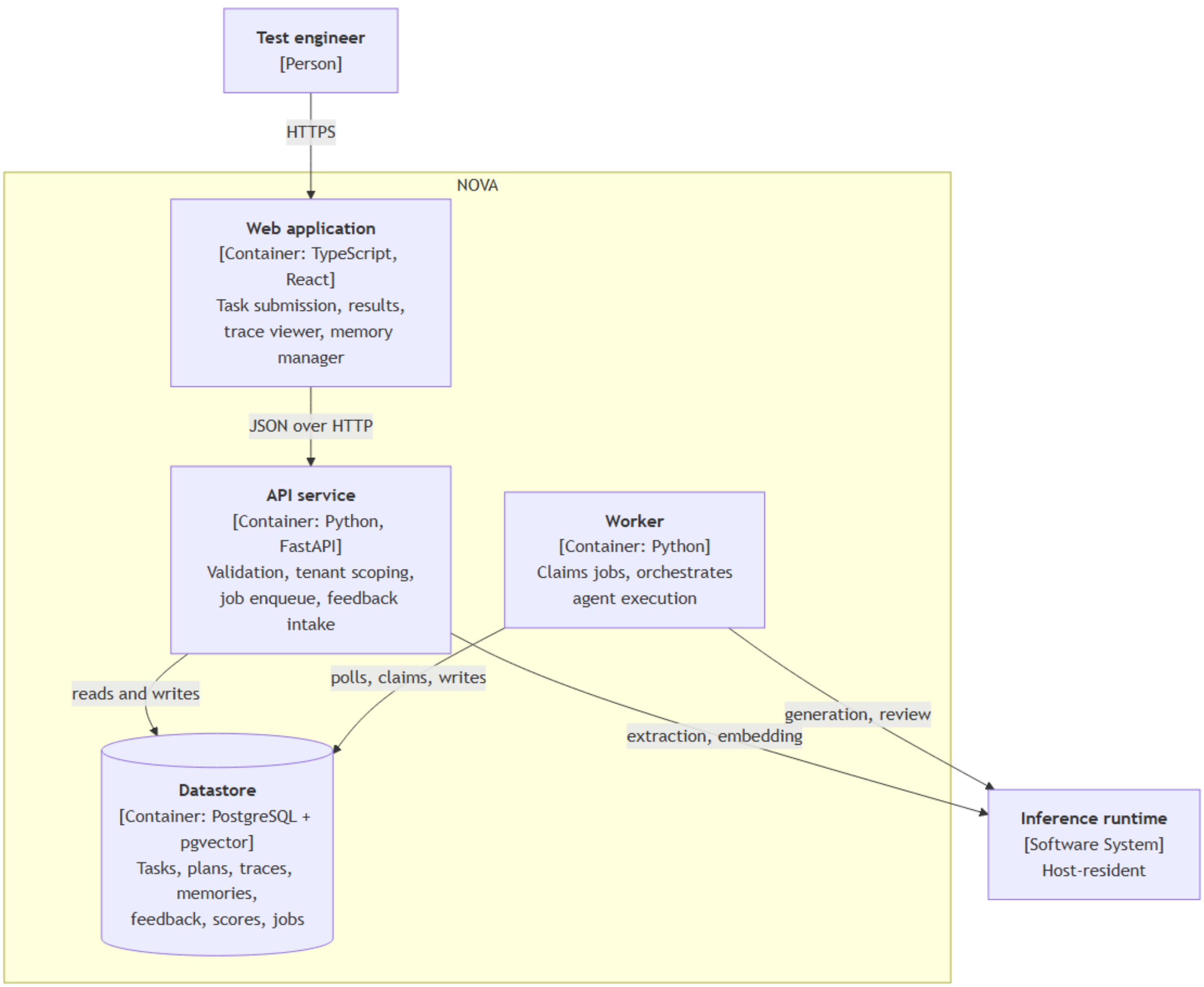
Quality goal	Approach	Rationale
Security of the memory path	Two independent controls in series: a mechanical validation screen, then mandatory human confirmation. Combined with a total absence of tool capability for the model	A screen against an open-ended attack surface will miss cases. A successful injection can produce a poor test case but cannot take an action, because no action exists. See ADR-0004
Analysability	Trace steps written incrementally, including retrieval scores for candidates that were evaluated and rejected	A trace persisted only on success is useless, because failure is when it is needed. Scores for non-selected candidates are what separate retrieval failure from generation failure
Testability	Deterministic checks first, model-based review confined to the residual. Evaluation dataset versioned by content hash. Configuration activation blocked at the data layer without a passing run	Model-based evaluation makes every metric dependent on a component whose reliability must itself be established
Confidentiality	Local inference by default. Provider changes require explicit user action. Logs carry identifiers and digests, never content	Automatic provider substitution would silently change where data goes

Quality goal	Approach	Rationale
Reliability	Durable job records claimed under row-level locks with lease expiry. Every retry bounded with a defined terminal state	An agent that retries without bound consumes a GPU and presents as a hang

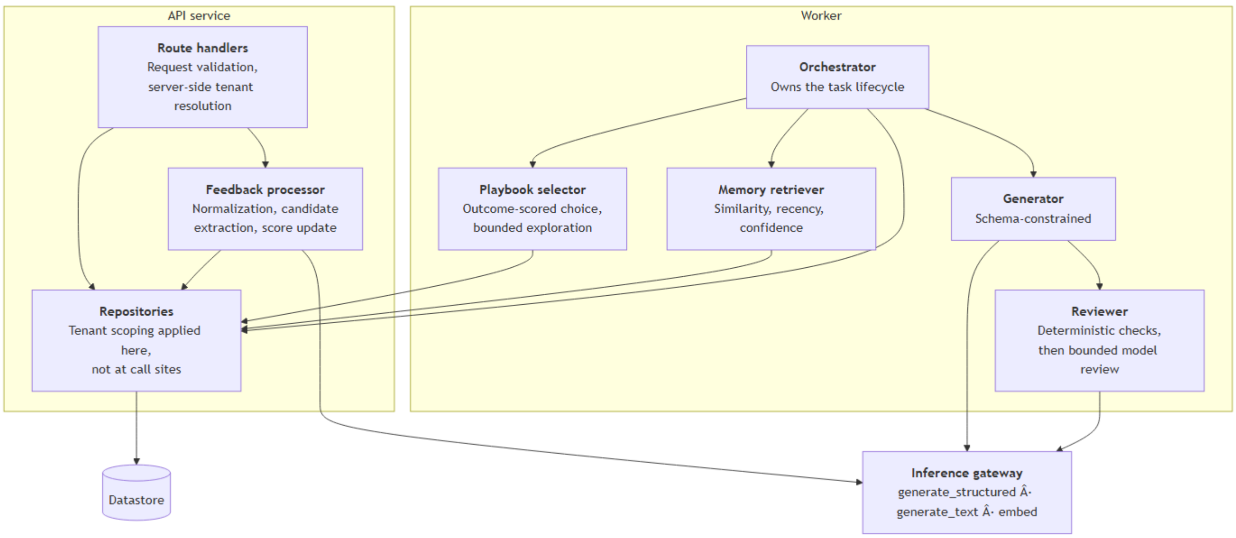
Technology selection follows from the constraints rather than preference: a single datastore holding both relational and vector data (ADR-0001), a job table rather than a broker (ADR-0002), and no agent framework (ADR-0003).

5. Building block view

5.1 Level 1: containers (C4 level 2)



5.2 Level 2: API service and worker components



5.3 Component responsibilities

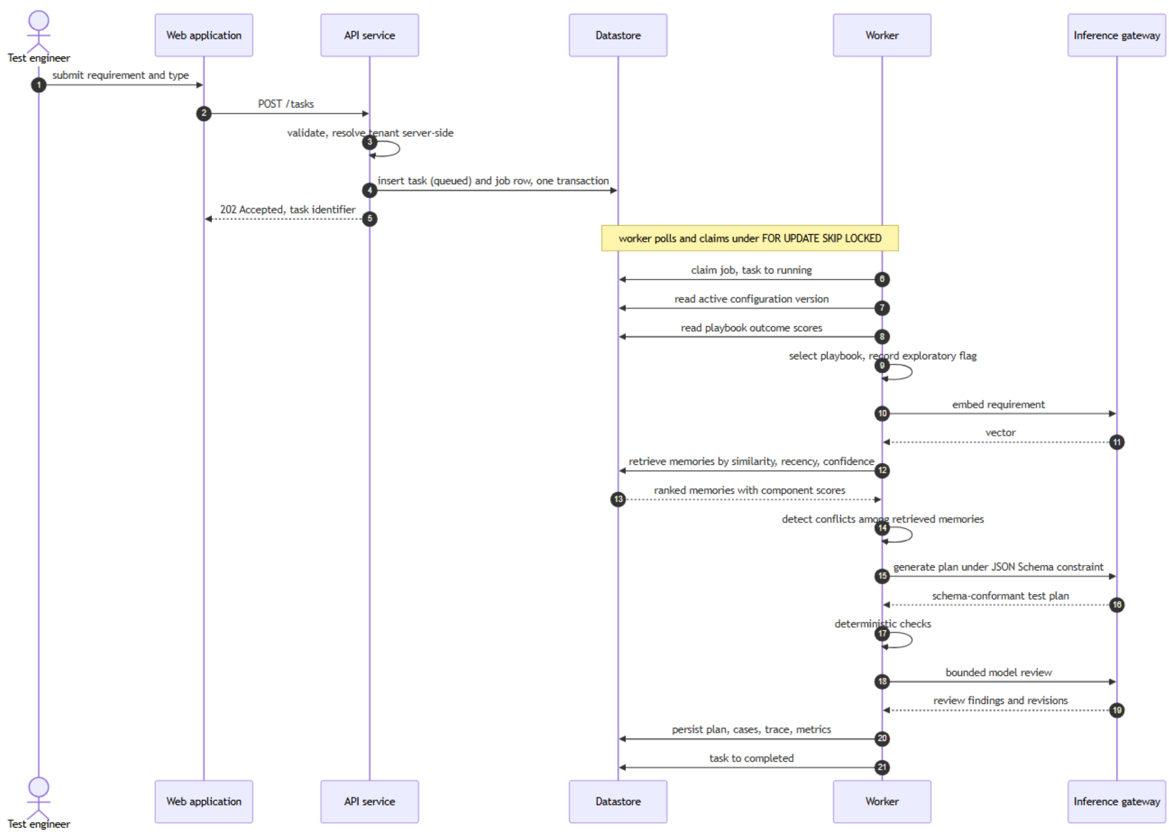
Component	Responsibility	Principal failure mode and response
Web application	Submission, results with per-case actions, trace viewer, memory manager, evaluation results	API unreachable, long task appearing stalled, large trace rendering slowly. Explicit error states, polling backoff, trace pagination. Renders untrusted content, so no unsanitized markup
Route handlers	Validation, server-side tenant resolution, job enqueue	Validation failure returns a structured error. Datastore failure returns 503 with no partial write
Repositories	All data access, tenant-scoped by construction	A query path that omits tenant scoping is prevented structurally rather than by convention
Feedback processor	Normalization, candidate extraction, score update	Extraction failure retains the raw correction rather than discarding it. Highest-risk path in the system
Orchestrator	Task lifecycle: select, retrieve, generate, check, review, persist	Any step failure records the failing step by name and persists the trace to that point
Playbook selector	Outcome-scored choice with bounded exploration	Cold start falls back to a static mapping. Bounded step size and clamping prevent a single event from changing the selection
Memory retriever	Ranked retrieval with explainable component scores	Embedding failure fails the task explicitly. A silent empty retrieval is the worst available outcome, because the system continues to appear healthy while no longer learning
Generator	Schema-conformant plan	Bounded repair retry with every attempt counted. Context pressure reduces retrieved memory

Component	Responsibility	Principal failure mode and response
	production	before requirement text
Reviewer	Deterministic checks, then bounded model review	The model review stage is optional by design. Deterministic verdicts return regardless, flagged partial. The pre-review plan is retained so any regression is visible
Inference gateway	Three operations over two provider implementations	Startup health check returns an actionable message. No silent provider substitution
Job store and worker loop	Durable asynchronous execution	Lease expiry permits reclamation after worker termination. Bounded attempts, then terminal failure. Partial trace always retained

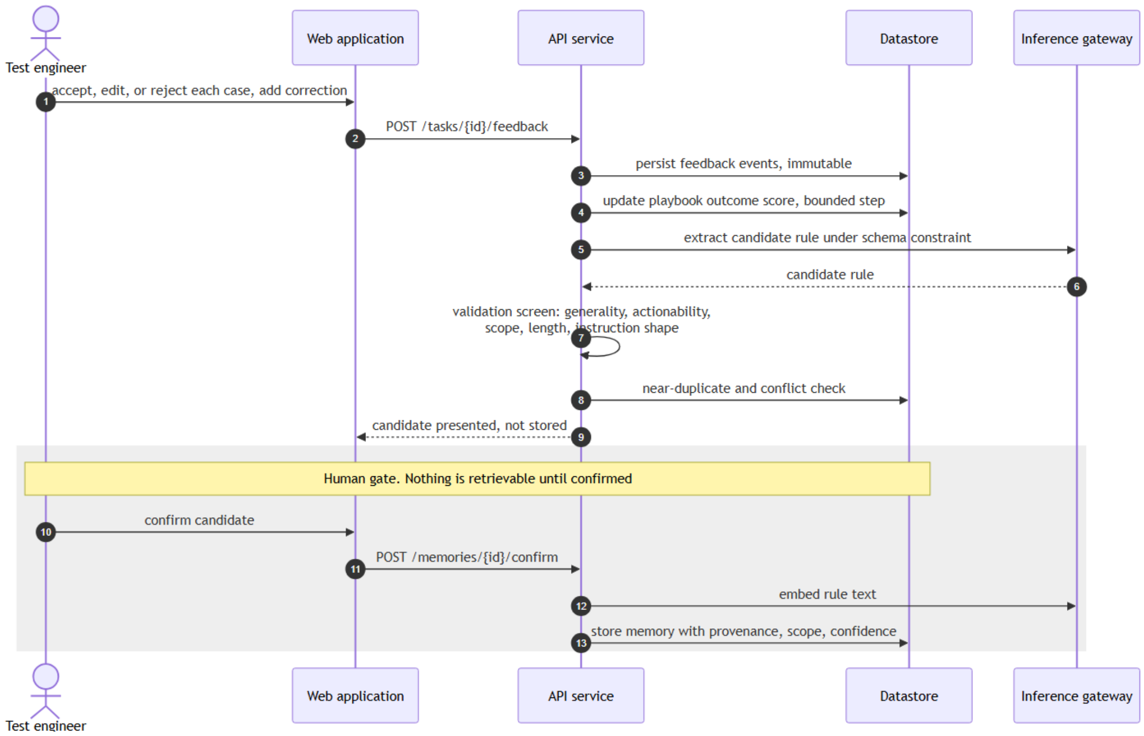
The gateway exposes three operations, not a general abstraction layer. Anything wider would be speculative.

6. Runtime view

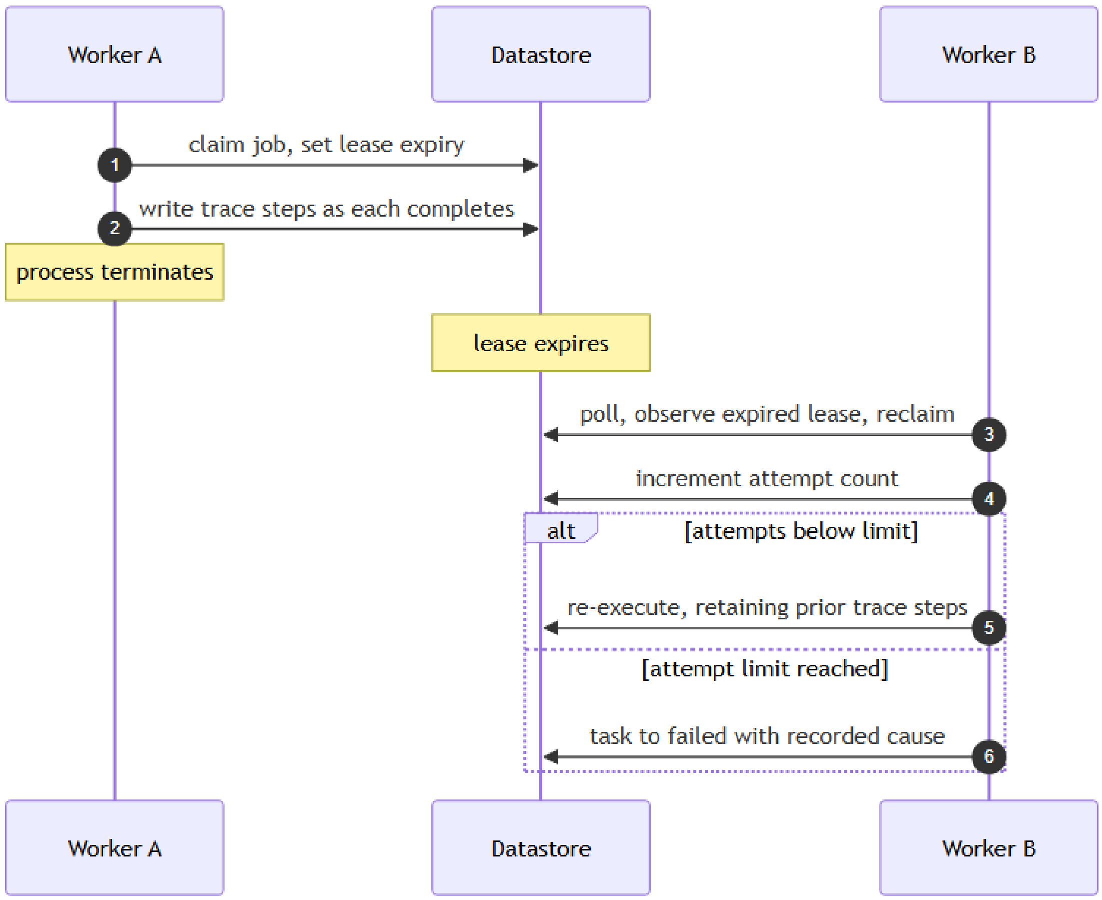
6.1 Nominal task execution



6.2 Correction to durable memory



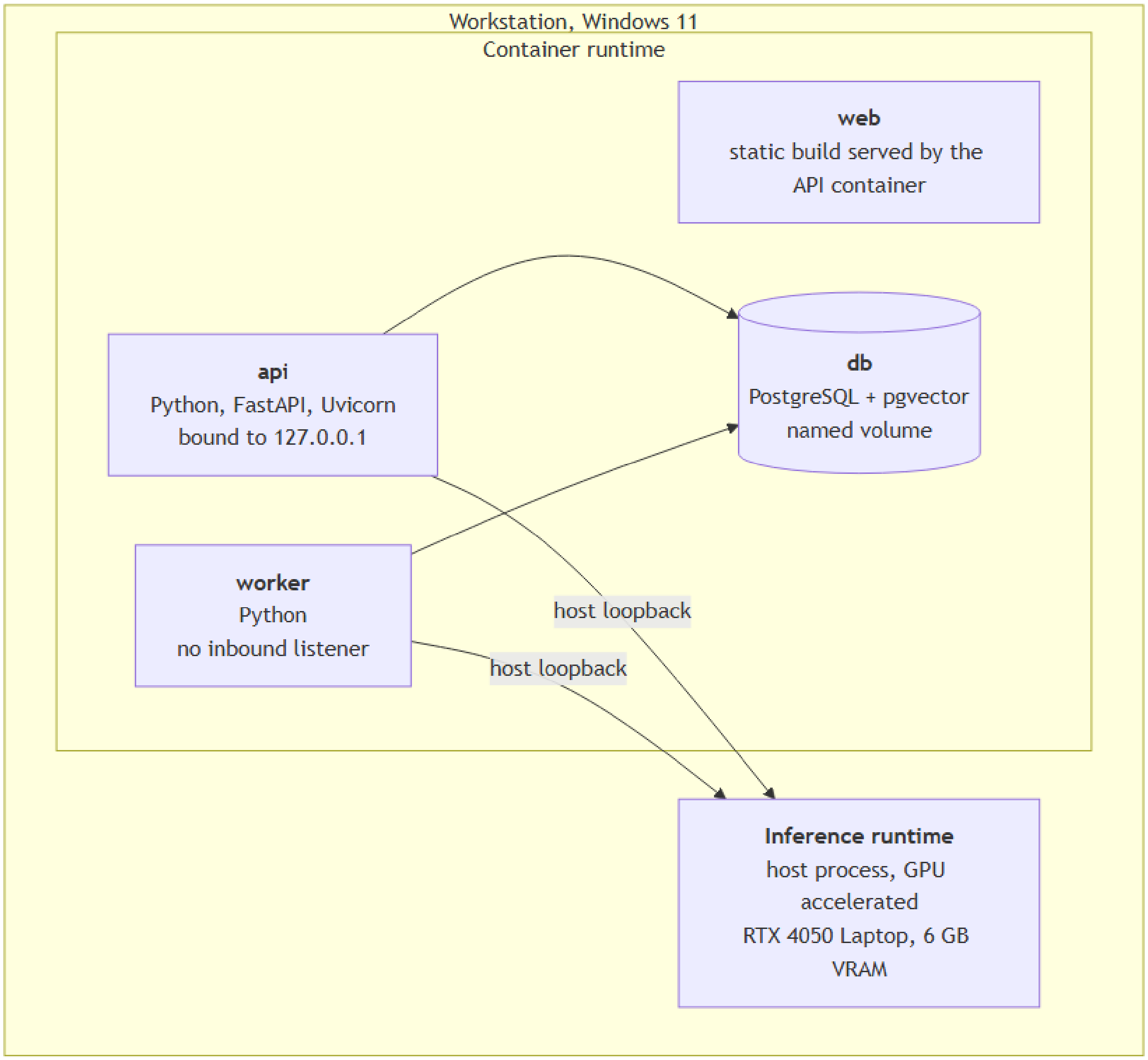
6.3 Worker termination and recovery



6.4 Synchronous and asynchronous boundaries

Operation	Mode	Rationale
Task submission	Synchronous, returns identifier	Execution takes tens of seconds
Task execution	Asynchronous worker	Durability and observable status
Status and result retrieval	Synchronous polling	Server-sent events only if polling proves inadequate
Feedback submission	Synchronous	Write-only, fast
Candidate extraction	Synchronous, within the feedback request	A single short call, and the user is awaiting the candidate
Embedding on confirmation	Synchronous	Sub-second locally
Evaluation runs	Asynchronous, offline	Minutes in duration. Never in a request path

7. Deployment view



Node	Rationale
Application containers	Reproducible startup from a clean checkout by a single command
Datastore container with a named volume	State survives container replacement. Backup is a documented dump and restore
Inference on the host, not containerized	GPU passthrough on this platform adds operational friction without benefit. The container-to-host path is verified: on Docker Desktop, `host.docker.internal` reaches the runtime on its default loopback binding, so no host-side binding change is required. A Linux-native engine has no equivalent host proxy and needs

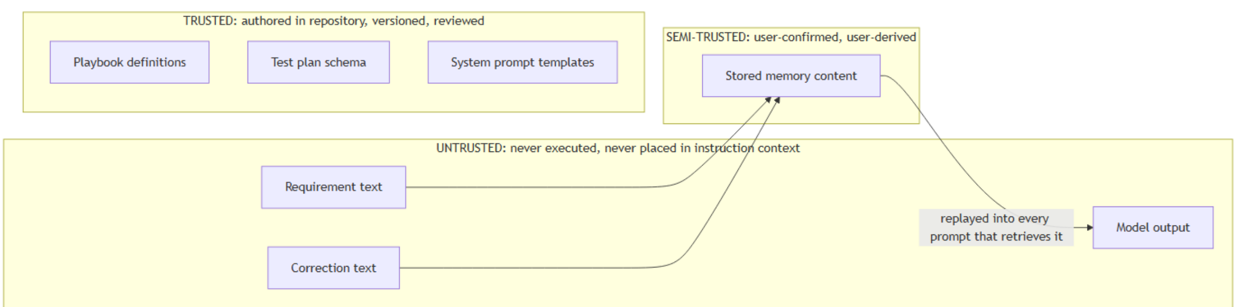
Node	Rationale
	`OLLAMA_HOST` widened with the port firewalled to the Docker subnet. NOVA-SPK-003

8. Crosscutting concepts

8.1 Domain model

Concept	Definition
Task	One execution of the requirement-to-plan workflow. The anchor for plan, trace, and feedback
Test plan	A summary and an ordered set of test cases, conforming to a versioned schema
Memory	A durable rule derived from a correction, carrying provenance, scope, confidence, and lifecycle state
Playbook	A named generation strategy: required case types, prompt fragment, check configuration. Versioned repository data, not code
Feedback event	An immutable record of one user judgment. The authoritative source from which scores are derived
Execution trace	The ordered record of every step, model call, and score for one task

8.2 Trust boundaries and security



Stored memory is a persistent injection vector. A single injected requirement affects one task. An injection surviving extraction and confirmation is replayed into every subsequent prompt that retrieves it, which is structurally equivalent to stored cross-site scripting.

Seven layers apply, none relied upon alone: structural separation of untrusted content, constrained decoding bounding the output shape, output validation, instruction-shape screening on candidates, human confirmation, absence of any tool surface, and an adversarial regression suite. The sixth is the strongest, and it derives from the architecture rather than from a filter.

Full analysis in NOVA-TM-001.

8.3 The learning mechanism

Adaptation occurs at five levels. None involves model weights.

Level	Mechanism	Persistence
L0	Working context, assembled per task	None
L1	Episodic records of tasks, outputs, outcomes	Database rows
L2	Semantic rules extracted from corrections	Database rows with provenance
L3	Scored retrieval over those rules	Query-time
L4	Outcome-scored playbook selection	Derived scores
L5	Evaluation-gated configuration change	Versioned records

Feedback normalization. Events map to a bounded score in [-1, +1]: unedited acceptance `+1.0`, minor edit `+0.5`, major edit `0.0`, rejection `-1.0`, user-added case `-0.5`, abandonment `-0.2`, export `+0.3`. The task outcome is the clamped mean of case-level contributions.

Three choices carry rationale. Clamping prevents a single large plan from dominating. A missed case is penalized less than an incorrect one, because omission is cheaper to remedy than a false positive. A major edit scores zero rather than negative, because the structure remained useful.

Implicit signals (abandonment, export) are weighted well below explicit feedback and never independently create or modify a memory. Abandonment is inherently ambiguous.

Retrieval. `score = w_sim Â· similarity + w_rec Â· recency + w_conf Â· confidence`, initially `0.6 / 0.2 / 0.2`. Scope and tenant apply as filters before ranking. Pinned in-scope memories return unconditionally. Component scores for every evaluated candidate, selected or not, are written to the trace.

Conflict handling. Two in-scope memories that are semantically similar but prescriptively opposed are surfaced, never merged. At confirmation the user chooses; at retrieval the higher product of confidence and recency prevails and the conflict is flagged. Silent averaging would make the memory store untrustworthy, because the user could no longer determine what the system holds to be true.

Strategy selection. Greedy over recorded outcome scores with probability `1 - Îµ`, uniform random otherwise, Îµ starting at 0.1 with a floor of 0.05. Updates are bounded and clamped, so no single feedback event alters a selection. Below a minimum observation count the static default prevails.

This is a bounded contextual bandit over four fixed options. It is not reinforcement learning and is not described as such. Four arms and a scalar reward is a modest mechanism, but it is real, scoreable, and explainable.

Outcome scores are derived, never authoritative, and are recomputable from the immutable feedback log. A poisoning incident or a change to the weighting scheme is therefore recoverable.

8.4 Persistence

A single datastore holds relational and vector data. The determining argument is correctness rather than performance: with separate stores, writing a memory requires two writes to two systems, and a partial failure yields either orphaned vector data or a memory that exists but can never be retrieved. The second fails silently and is indistinguishable from a system that has stopped learning. Within one transaction the failure mode does not arise.

The embedding dimension is fixed at schema creation. Changing the embedding model requires re-embedding every memory and rebuilding the index. This is documented rather than discovered, and is the reason the embedding decision was treated as blocking. It is settled: `embeddinggemma` at 768 dimensions, per ADR-0005 on the evidence in NOVA-SPK-002.

8.5 Asynchronous execution

Jobs are rows claimed under `SELECT ... FOR UPDATE SKIP LOCKED` with a lease that expires. Because the job row shares a transaction with the task record, no window exists in which a task is present without its job. Diagnosis of a stalled job is a query rather than an operational investigation.

8.6 Error handling

Errors are returned in RFC 9457 problem-detail form, giving one shape for every failure and one rendering component. Every retry loop is bounded with a defined terminal state.

Trace steps are written as they complete, so a task failing at generation still exposes its playbook selection and retrieval results.

8.7 Observability

Spans correlate across API, worker, and inference calls. Logs carry identifiers and prompt digests, never requirement text or memory content, and an automated assertion enforces this. Content logging exists only as an explicit debug mode. Logging failure degrades to standard output and never blocks task execution.

Content-free logging is not incidental. Local inference is the confidentiality property, and logs are the most probable route by which it would be undermined.

8.8 Configuration and versioning

Prompt templates, retrieval weights, exploration rate, and model configuration are held as versioned records. A version reaches active status only with an associated passing evaluation run, enforced as a data-layer constraint rather than an interface check.

Every task pins the configuration version under which it executed. Without that pin, historical results are uninterpretable and regression comparison is meaningless.

8.9 Testability

Deterministic checks precede model-based review. Integration tests execute against a real datastore and a fake inference provider, because they exist to verify wiring, and model behaviour is the evaluation layer's concern. Both provider implementations satisfy a shared contract suite.

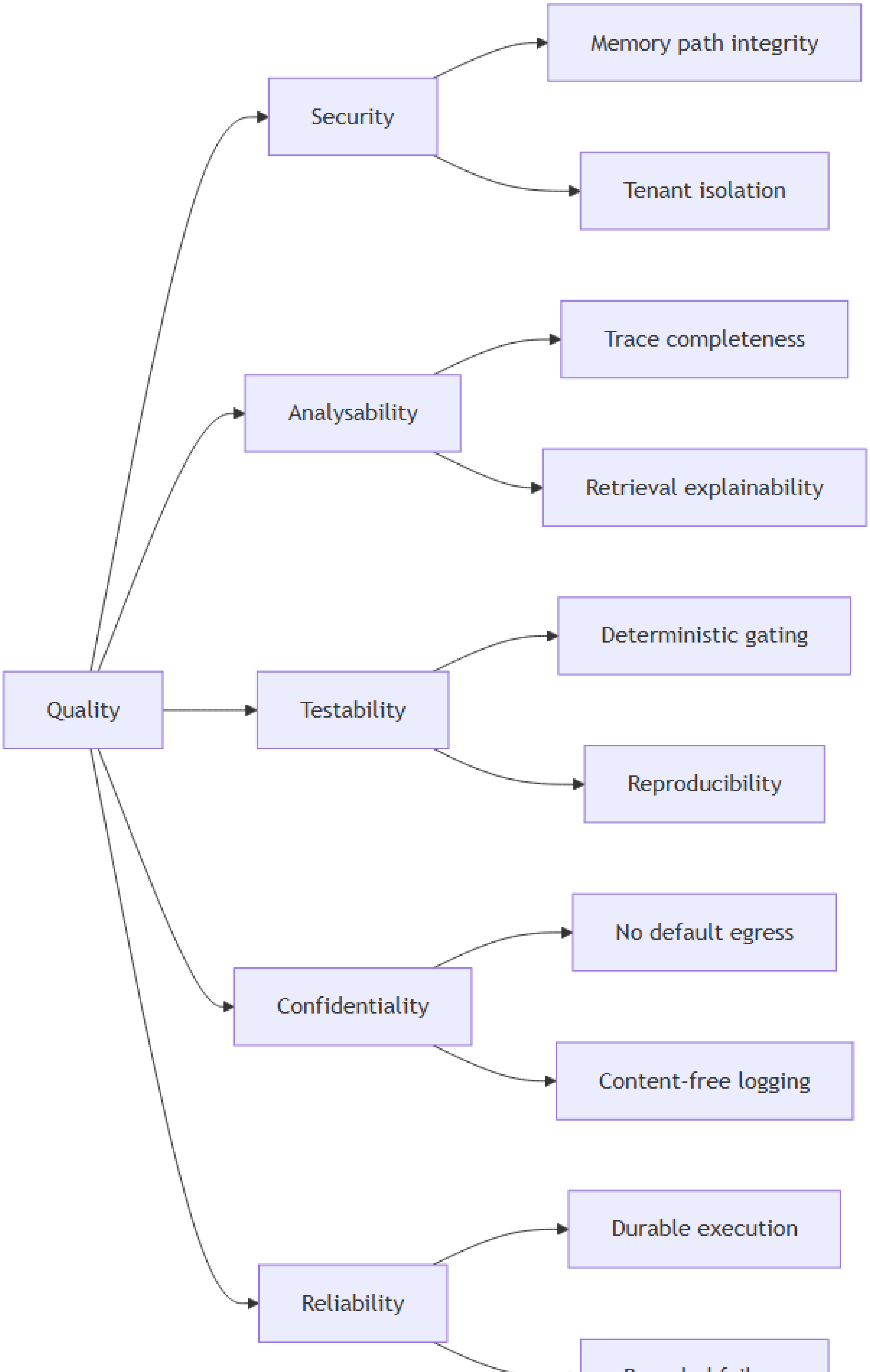
9. Architecture decisions

Significant decisions are recorded as ADRs in MADR 4.0 format. Remaining decisions are held in NOVA-DL-001.

ADR	Decision	Status
0001	PostgreSQL with pgvector as the sole datastore	Accepted
0002	Job table and polling worker rather than a message broker	Accepted
0003	No agent framework	Accepted
0004	Mandatory human confirmation for memory writes	Accepted
0005	Embedding model and vector dimension	Proposed

10. Quality requirements

10.1 Quality tree


Quality
Security
Analysability
Testability
Confidentiality
Reliability
Memory path integrity
Tenant isolation
Trace completeness
Retrieval explainability
Deterministic gating
Reproducibility
No default egress
Content-free logging
Durable execution

10.2 Quality scenarios

ID	Scenario	Stimulus	Required response
QS-1	A correction is submitted whose text is shaped as a system instruction	Malicious or accidental input	Screened and flagged. Never stored without explicit user action on the flagged item
QS-2	Retrieval is performed for one tenant while another holds highly similar memories	Query execution	Zero records from the other tenant. Verified as an unconditional gate
QS-3	A task fails during generation	Provider timeout	Playbook selection and retrieval results remain inspectable in the persisted trace
QS-4	Output omits an expected test case	User review	The trace distinguishes a memory that was never retrieved from one retrieved and not applied
QS-5	A prompt change degrades output quality	Configuration change submitted	The regression gate blocks activation before merge
QS-6	A prior execution must be reproduced	Same input, seed, model, configuration version	Equivalent output
QS-7	A hosted provider becomes selected	Configuration action	Only through explicit user action. Never as a fallback. Visible in the interface
QS-8	The worker process terminates mid-task	Process failure	The job is reclaimed after lease expiry. The partial trace survives
QS-9	The inference runtime is unavailable	Dependency failure	Submission still succeeds and queues. Readiness reports unhealthy. The error names the required action

11. Risks and technical debt

ID	Risk or debt	Severity	Response
R-01	Correction extraction may not reach usable precision on a local model	High	Measured at a defined checkpoint before dependent interface work. Fallback is user-authored rules with model assistance
R-03	Inference nondeterminism may exceed evaluation gate margins	High	Variance measured before any threshold is set. Dataset grows rather than thresholds loosening
R-05	The Correction Recurrence Rate depends on a structural similarity	Medium	Threshold fixed from measured distributions, documented, held constant, and always

ID	Risk or debt	Severity	Response
	threshold that is difficult to defend		reported alongside retrieval precision
~~R-09~~	~~Container-to-host inference reachability is unverified on this platform~~	Closed	**Resolved 2026-09-09.** Reachable with the inference runtime on its default loopback binding, through Docker Desktop's host proxy. NOVA-SPK-003
R-11	Persistent injection defences are unproven	Medium	The adversarial suite establishes a floor, not a guarantee, and is described as such
R-13	The model review stage may not justify its latency	Low	Designed to be removable. Measured before and after
R-16	Introducing any tool capability would invalidate the security model	Low	Recorded so that the consequence is not overlooked. Such a change requires rewriting NOVA-TM-001 rather than amending it
D-01	Trace volume has no pruning policy	Low	Acceptable at single-user scale. Named so it is not discovered later
D-02	Memory conflict resolution is manual	Low	Does not scale beyond a moderate memory count. Acceptable at expected volume

Full register: NOVA-RR-001.

12. Glossary

See NOVA-SRS-001 section 1.4.

Revision history

Version	Date	Author	Change
0.1	2026-09-08	Laxmi Poudel	Initial draft
0.2	2026-09-08	Laxmi Poudel	Embedding dimension fixed at 768 in section 8.4. R-04 moved to closed
0.3	2026-09-08	Laxmi Poudel	R-08 moved to closed
0.4	2026-09-09	Laxmi Poudel	R-09 moved to closed. Section 7 records the verified container-to-host path and the Linux-engine caveat